Which Target to Focus on: Class-Perception for Semantic Segmentation of Remote Sensing

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***Abstract*— Deep-learning-based (DL) methods have dominated the task of semantic segmentation of remote sensing images. However, the sizes of different objects vary widely, and there is a great deal of label noise due to the inevitable shadows. Therefore, there is an urgent need for a method that can precisely handle complex ground data. In this article, we propose an interclass enhanced network (ICEN) for representing features of varying sizes. It comprises two branches: sparse representation network (SPN) and feature extraction network (FEN). Then, a class-perception block (CPB) is inserted between the two branches to instruct the SPN's low-level semantic features to be merged into the deeper network. Such a block can reduce label noise in remote sensing image segmentation. In addition, the proposed EIRI provides a more precise classification process for target edges containing many misclassified points without requiring excessive computational overhead. The experimental results of our proposed class-perception network (C-PNet) achieve competitive performance on the Vaihingen, Potsdam, LoveDA, and UAVid datasets.**

***Index Terms*— Class-perception, computational overhead, label noise, multiresolution, remote sensing, sparse representation.**

I. Introduction

SEMANTIC segmentation of remote sensing images plays an important role in urban planning [1], land use [2], and disaster management [3]. As an important semantic segmentation task, it is performed by manipulating the input raw image to obtain the class to which each pixel belongs. Through the semantic segmentation of remote sensing images, industrial and related organizations can have a detailed observation of the conditions on the ground and take timely and proactive actions, such as the management of traffic [4] and the renewal of maps [5]. With the continuous advancement of Earth exploration technique in recent years, many high spatial resolution remote sensing images are widely available [6]. Therefore, it is more critical than ever to find an efficient automatic semantic segmentation method to exploit the abundant contextual information and interclass relationships in remote sensing images.

For remote sensing images, classification algorithms extract crucial information from images by analyzing statistical features such as gray values, colors, and gradient values within the image. However, these conventional methods, which are manually designed, lack the ability to capture high-level semantic features in images, such as contextual information [7], [8], [9]. With the advancements in imaging techniques for remotely sensed images, the depiction of ground targets has become more refined and sophisticated [6]. To harness the power of image semantics and extract meaningful features, deep learning (DL) methods with autonomous understanding have taken the spotlight [10], [11], [12].

Supervised learning [1], [10], [13] dominates the remote sensing image segmentation algorithm based on DL, which requires sufficient finely labeled samples to make the network learn to perceive the semantical content in the images and pixel-by-pixel predict the labeled image [14], [15], [16], [17]. Since labeling each target is a time-consuming and resource-consuming task, and the error of land cover labeling will significantly reduce the performance of the deep network [18], [19], [20], [21]. Although weakly supervised ones demand less elaborate labels, recent research on the field always faces unsatisfactory performance and high computational complexity problems compared with the supervised learning method [22], [23].

When implementing the above deep neural network strategies on the cutting edge to perform remote sensing image segmentation, the heterogeneous ground targets are presenting more challenges for researchers as follows.

1) The sizes of different categories of targets in remote sensing images vary greatly, even not in the same order of magnitude. For example, a building's size can reach

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500 × 500 pixels than a small car, which is usually only 30 × 30 pixels. It is unreasonable to use a uniform network architecture to segment these two targets with such a considerable size difference.

2) The prevalence of shadows in remote sensing images makes a part of the training dataset have labeling errors, thus bringing the problem of label noise. Reducing the negative effect of label noise and making the model more robust is a pressing challenge.
3) The amount of different ground targets coexisting in a single image, averagely allocating limited computational complexity to every pixel, produces counterproductive unbalanced results. Because that particular small targets in the urban area, such as cars, deserve more calculation on every pixel as the ambient surroundings are of great complexity. On the other hand, careful observation of each pixel in a large forest area would waste too much time and resources. Therefore, adaptively adjusting the allocation of computational resources according to different objectives can save unnecessary resource consumption and improve learning efficiency.

To solve the above three issues, we propose class-perception network (C-PNet), the details of which are presented in Section III, with three main contributions as follows.

1) We propose a cooperative network architecture containing two branches, feature extraction network (FEN) and sparse representation network (SPN). FEN is used to extract in-depth semantic features and focuses on solving the segmentation problem of large-size targets. SPN is used to extract shallow semantic features and tends to segment small-size targets.
2) Between FEN and SPN, we introduce class-perception block (CPB), which uses the in-depth semantic features in FEN to control whether SPN's shallow semantic features are fused into SPN's deeper network.
3) Entropy is proposed here to generate a levity map that measures the uncertainty of the neural network's predicted value for each point. According to the levity map, we bias the network's computational load to those still not well considered by the neural network. A multiscale fusion is used to reconsider the points with more abundant features.

II. Related Work

A. Shallow Feature Description

Shallow features tend to have higher spatial resolution and retain more texture detail features. As a result, they are playing an important role in remote sensing image segmentation. Before the advent of the DL era, researchers used many arithmetic-based methods for image feature extraction, which contributed a lot to representing low-level features in images. For example, the edge detection algorithm proposed in [24] and [25], the direction feature representation proposed in [26], and the local feature extraction in [27]. However, these handcrafted feature extraction methods based on mathematics are usually vulnerable to noise. Therefore, DL is attracting increasing attention with the outstanding ability of self-learned features.

Of the recent research results on in-depth learning-based semantic segmentation, it is feasible to extract high-level semantic features that are difficult to obtain by mathematical methods [7], [9]. However, many shallow-level features are lost by merely exploiting high-level features for semantic segmentation [14]. In particular, when smaller targets are present in an image, they are usually overlooked because of the fewer pixels they occupy. However, these targets are likely to be the target for segmentation [28], [29], [30]. Therefore, these low-level features, which are weakened in deep networks, play an important role in depicting shallow features of images.

Therefore, researchers have paid more attention ever to the extraction of low-level semantic features. FCN serves as the cornerstone of image segmentation methods based on deep convolutional neural networks, paving the way for the fusion of high-level and low-level semantic features. In subsequent semantic segmentation methods with this encoding–decoding architecture, low-level features are usually processed and are added during the composition of the output results [10], [14], [16]. In [31], for targets of different sizes that are in the same image, the algorithm uses features of different depths for prediction. In the segmentation network architecture based on RPN [15], the grid's size also affects the effectiveness of the instance segmentation. For instance, in the RPN-based network architecture, segmentation [15], the grid's size also affects effectiveness. Kirillov et al. [32] treat the image segmentation problem as a computer rendering task, which adaptively samples for higher level feature maps and complements them with lower level features. Chan et al. [33] argue for efficient capture of low-level edge information by replacing the traditional convolution kernel with PCA convolution.

In remote sensing image segmentation, the importance of low-level semantic features will be more prominent due to ground targets with widely varying sizes when comparing, for example, a small car with a large forest. Guided by high-level features solely, these targets are likely to be overlooked. Although the number of pixels is far different between the two, the classification effect has the same weight under the average accuracy (AA) metric. Traditional machine-learning-based methods for remote sensing image feature extraction are usually realized using MRF [34] and SVM [35].

However, compared with the shallow features automatically extracted by deep neural networks, the features they extract lack specificity. Therefore, in the field of remote sensing image segmentation in recent years, academics have gone further and further in exploring deep neural networks for shallow feature extraction and utilization. In [13], to exploit shallow local low-level semantic features, a deep network is used to learn them exclusively. [29] used the concept of residual learning to improve an FCN for the segmentation of small-scaled cars in remote sensing images, enabling the network to learn contextual semantic information from feature maps of different depths. In [36], they adaptively enhance the different depth of features for targets of different sizes during multiscale fusion, which means that when dealing with small targets, enhancement is done for low-level features and conversely

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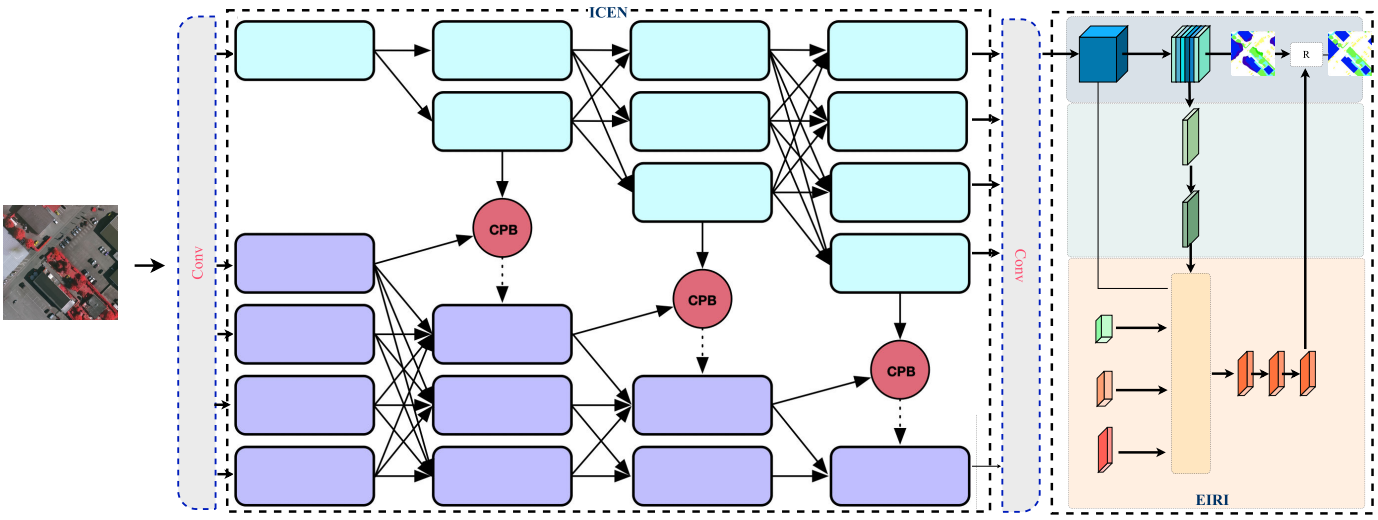


Fig. 1. Central part of C-PNet, but upsampling and 3 × 3 convolution are not included. Upper triangle in ICEN is an FEN, and lower triangle is an SPN. When low-resolution feature maps are obtained using a 3 × 3 convolution, upper triangle controls whether the lower resolution feature maps in lower triangle are fused to the higher ones via a gate function. The details are shown in Section III.

for deep-level features of large-size targets. In this way, the shallow feature information is retained.

B. Multiscale Fusion

Given that low-level features contain more location and detail information with lower semanticity. For deep-level features, they face the problem of shallow resolution, although they contain high-level semantic features. Therefore, it is extremely demanding to design a feature fusion method that can efficiently combine both and thus make the image's feature information more comprehensive as soon as possible. Concat [37], [38], [39], [40], [41] and add [9] are the two most common ways of early fusion. The fused results train the classifier and get the final result by performing feature fusion before getting the final result. However, simply concatenating feature maps can lead to various problems, such as large spatial resolution differences and the mean value of feature maps at different scales. Therefore, they are usually normalized before concatenation.

Zhao et al. [38] perform L2 regularization for each layer before fusion to ensure that the fused features are not biased toward a certain layer. The counterpart of early fusion is late fusion. FPN [42] is a representative of late fusion. A classifier is trained to obtain multiple classification results at different sizes by a featured image pyramid for different size features. The final output is optimized by combining the different results. Inspired by FPN, YOLO V3 [17] improved the detection of small targets by detecting independently on the fused feature maps at multiple scales with the help of the architecture of FPN. Multiscale fusion becomes important in remote sensing image segmentation.

Paisitkriangkrai et al. [43] fused hand-designed features with semantic features extracted by deep neural networks to improve further the accuracy of pixel-level classification. Ding et al. [13] designed a module for fusing different feature maps obtained from different convolutional layers to achieve a sharp prediction for each point. Furthermore, [44] designed a module for adaptively selecting features for fusion based on the semantic information required by the target. Liu et al. [36] proposed a method that can autonomously perform feature fusion based on the size of the perceptual field with the help of the idea of nonlocal network [45].

III. Class-Perception Network

In this section, we introduce the various details of our proposed network. The overall architecture of our method is shown in Fig. 1. In this network, to solve losing features after multilayer convolution of small-sized targets, we propose an interclass enhanced joint network (ICENet), which contains two branches. They are introduced in Section III-A. To reduce the influence of a large amount of label noise in remote sensing images, we use the CPB between two branches, which can be seen in Section III-B. Finally, to balance the computational overhead within the targets at the boundary, we propose a method called EIRI, shown in Section III-C in detail.

A. Interclass Enhanced Network (ICEN)

At present, semantic segmentation of natural images has achieved very high accuracy, while many excellent network architectures have emerged, such as HRNet, DeepLab, and EfficientNet. Simultaneously, many backbones, such as AlexNet, GoogLeNet, ResNet, and its variants ResNeXt, Res2Net, and ResNeSt, also appeared successfully. Therefore, a classic idea is to migrate the network architectures, which are better achieved in natural images, processing remote sensing images.

However, the size of different ground targets in remote sensing images varies greatly, and even the size of the target is not the same order of magnitude. For example, in the ISPRS dataset, the image's size is about 2000 × 1500 pixels. Wherein the size of buildings can reach 500 × 500 pixels in some

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cases. However, in contrast, the size of cars is only about 30 × 30 pixels. In this case, after the same number of convolutions as the larger sized ones, the smaller sized target will retain very few features, and thus the gradient disappears. In this case, it is unreasonable to use a uniform and consistent network to segment the two kinds of ground targets.

Inspired by the above challenges, HRNet retains the high-resolution branches from the beginning to the end. As the number of layers increases, the model continuously downsamples to obtain low-resolution branches and compensates for the loss of lost high-resolution features by increasing the number of channels. In this way, the network's feature maps fuse the features at high and low resolutions, guaranteeing the segmentation accuracy of large-sized targets at high resolution. However, HRNet does not consider the segmentation problem of small-sized targets, prevalent in remote sensing images. As a result, to solve the problem of large differences in target sizes, we significantly enhance the architecture of HRNet and propose the ICENet, which contains two branches, SPN and FEN.

The upper triangular part of Fig. 1 shows the network architecture of FEN for representing and capturing the in-depth semantic features in images. Meanwhile, downsampling is required whenever a new low-resolution branch is generated, so we call it FEN, which is used for large-size targets. Specifically, we use HRNet as the FEN in our experiments. The lower triangular part of Fig. 1 shows the network architecture of SPN. To retain as many shallow features as possible for small-size targets, the input of SPN is divided into four branches with different resolutions instead of only one high-resolution branch contained in the FEN. Simultaneously, to avoid losing the original features of small-sized targets after multilayer convolution, the number of convolutional layers in all the branches in SPN is reduced to half of that in FEN. Therefore, we call it SPN, which focuses on solving the segmentation for small-size targets. In conclusion, the differences between FEN and SPN are described as following.

1) FEN is a dense network used to extract high-level features of the ground target, such as the texture and the relationship between different targets; SPN is a sparse network used to retain low-level semantic features of the ground target, such as the shape and color.
2) FEN has more convolutional layers, which are used to segment large-size targets; SPN has fewer convolutional layers, which are used to segment small-size targets, such as cars.
3) FEN and SPN concatenate the feature of two branches after a certain number of convolutions. On one hand, the low-level features of the targets can be retained, and the high-level features of the targets can be deeply represented. On the other hand, the segmentation accuracy of the large target can be guaranteed, and the semantic features of the small target can be avoided to be lost after being convolved by multilayer.

FEN and SPN form the main framework of ICENet, which makes the network better applicable to the situation of remote sensing images with huge differences in targets. However, there are still many problems in remote sensing images, such as label noise. For this reason, we propose CPB to make FEN and SPN work collaboratively.

Algorithm 1 Train Process of Class-Perception Block

Input: The feature map from FEN, $\mathcal{F}_n$; the number of boundary points, k; the information threshold, ϱ; the number of eigenvalues, t;

Output: The flag of whether low-level semantic features are fused, τ; the transformation matrix, T_0;

Use **Class-Perception Block** to obtain the ***Top-k*** points on the feature map $\mathcal{F}_n$, $(x_1, y_1), (x_2, y_2), \ldots, (x_k, y_k)$;
Normalize k points to $(x_1', y_1'), (x_2', y_2'), \ldots, (x_k', y_k')$, where T_0 is transformation matrix;
Extract the t most important eigenvalue, $\lambda_1, \lambda_2, \ldots, \lambda_t$;
Calculate τ: $\tau = \delta(\varrho - \sum_{n=1}^{t} \lambda_n)$;
return τ, T_0;

B. Class-Perception Block

ICENet solves the problem of small target feature loss. However, the two branches in ICENet do not have any weight sharing during the training process. Therefore, they cannot share their feature during this process. This limits the mutual integration and promotion of high-level and low-level semantic features. In the meantime, a few label errors which exist in the remote sensing images make themselves contain a certain amount of label noise. The more shallow the network, and the more the low-level semantic features, the more they are affected by the label noise.

To solve the above two problems, we introduce a CPB between the two branches of ICENet and use the high-level semantic features in FEN to control whether the low-level semantic features in SPN are fused into the deeper network in SPN. As we all know, if a fixed size region of an image contains more edge features, the richer information this region contains, so it is also indispensable to fuse some low-level semantic features, such as shapes and contours. Since the edges of the target can be roughly detected using the differential gradient of the image, an intuitive idea of CPB is that for a pixel (x, y) in the image and its feature value $f(x, y)$, if it satisfies

$$\|\nabla f\| = \sqrt{\left(\frac{\vartheta f}{\vartheta x}\right)^2 + \left(\frac{\vartheta f}{\vartheta y}\right)^2} > \varrho \tag{1}$$

we fuse the low-level semantic features in SPN into the deeper network, where ϱ is a dynamic threshold. Using differential gradient can solve the problem of low-level feature fusion to some extent.

However, the distribution of features in remote sensing images is much more complex than that of natural images and contains more label noise. This method only gives a fuzzy direction to solve the problem, but it cannot meet the demand for accuracy in semantic segmentation of remote sensing images. Therefore, we propose a method called CPB, and the details of the train process are shown in Algorithm 1: where T_0 is used for test process.

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The core idea of Algorithm 1 is to downscale the feature maps from FEN and use a small computational overhead to extract important features in the feature maps, such as texture information in the images, while removing unimportant features, such as noise. Using it to calculate the eigenvalues for $\boldsymbol{k}$ points with the largest responses in the FEN, the t largest of them is selected, $\lambda_1, \lambda_2, \ldots, \lambda_t$. The larger $\sum_{n=1}^{t} \lambda_n$ is, the less the boundary information is contained in the region where the current feature map is located. Therefore, the low-level semantic features in SPN are discarded at this time. On the contrary, if $\sum_{n=1}^{t} \lambda_n$ is smaller, the importance of the boundary information contained in the current region is similar, indicating that the current region is richer in information, and the low-level semantic features in SPN are fused into the deeper network at this time.

In the test process, there is no real boundary information for backpropagation. Therefore, the simple control of whether to fuse low-level features in the SPN using the method of Algorithm 1 seems insufficient. For this reason, we search for possible edge contours using the grayscale normal line direction of k points. This process is repeated until the maximum number of iterations. If the algorithm does not converge, the current region contains insignificant edge features and τ is false; otherwise, τ is true.

The coordinates of k points are obtained using Algorithm 1, $[(x_{ij}, y_{ij}), j = 1, 2, \ldots, k]$, and they are expanded into s 1-D vectors

$$\boldsymbol{x_i} = (x_{i1}, y_{i1}, x_{i2}, y_{i2}, \ldots, x_{ik}, y_{ik})^{\mathrm{T}}, \quad i = 1, 2, \ldots, s \tag{2}$$

and the average shape of k points is defined as follows:

$$\bar{\boldsymbol{x}} = \frac{1}{s} \sum_{i=1}^{s} \boldsymbol{x_i}. \tag{3}$$

Therefore, the covariance of k points is

$$\boldsymbol{S} = \frac{1}{s-1} \sum_{i=1}^{s} (\boldsymbol{x_i} - \bar{\boldsymbol{x}})(\boldsymbol{x_i} - \bar{\boldsymbol{x}})^{\mathrm{T}}. \tag{4}$$

To be able to fit the contour of the feature map from FEN, we use the weighted sum of each eigenvector to approximate the normalized test process

$$\boldsymbol{x} = \bar{\boldsymbol{x}} + \boldsymbol{\Phi b} \tag{5}$$

where $\boldsymbol{\Phi}$ is the set of eigenvectors corresponding to the t largest eigenvalues, and $\boldsymbol{b}$ is the weighted vector. Therefore, we have

$$\boldsymbol{b} = \boldsymbol{\Phi}^{-1}(\boldsymbol{x} - \bar{\boldsymbol{x}}). \tag{6}$$

$\boldsymbol{\Phi}$ is a symmetric matrix, so $\boldsymbol{\Phi}^{-1} = \boldsymbol{\Phi}^{\mathrm{T}}$. Thus,

$$\boldsymbol{b} = \boldsymbol{\Phi}^{\mathrm{T}}(\boldsymbol{x} - \bar{\boldsymbol{x}}). \tag{7}$$

Therefore, we can summarize the test method based on CPB, and the details of the test process are shown in Algorithm 2.

The update strategy of k key points added during testing motivates the model to accurately control whether to fuse low-level semantic features or not. In the experiments, according to the forward order of the network, the maximum number of iterations n is taken as 4, 8, 12, and 16 in order.

Algorithm 2 Test Process of Class-Perception Block

Input: The weighted vector, $\boldsymbol{b}$; the transformation matrix, $\boldsymbol{T}_0$; the maximum number of iterations, n;

Output: The flag of whether low-level semantic features are fused, $\boldsymbol{\tau}$;

Initialize $\boldsymbol{b}$, $\boldsymbol{b} = 0$;
Use 1 to obtain k points, get a rough boundary;
Calculate the inverse matrix of $\boldsymbol{T}_0$, $\boldsymbol{T}_0^{-1}$;
for $num = 1$; $num < n$; $num++$ **do**
    Search for a cross section of length $2k + 1$ in the direction of the gray normal line at each sampling point;
    Use the position of the point in the middle of the section as the new boundary point;
    Use 2 to normalize every point, get $\boldsymbol{T}$;
    Use $\boldsymbol{b} = \boldsymbol{\Phi}^{\mathrm{T}}(\boldsymbol{x} - \bar{\boldsymbol{x}})$ to update $\boldsymbol{b}$;
end for
Use Algorithm 1 to calculate τ;
return τ;

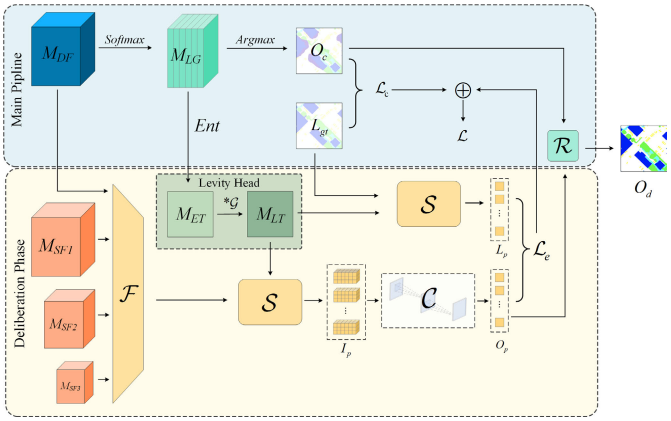


Fig. 2. Process of the phase of EIRI. The blue, green, and orange rectangles contain the process of the three step correspondingly, namely, main pipeline, LH, and DP.

C. Entropy Instructed Resource Inclination (EIRI)

In the field of semantic segmentation, it is common to use DL methods that require the same amount of computation for each pixel. In remote sensing images, however, the amount of computation required for each pixel can be very different in many cases. Therefore, the concept of entropy is harnessed for finding the points that need to be further investigated which is infested at the edge of the targets. After deliberately considered, the output yielded a more satisfactory performance particularly on the contiguous part. The process of EIRI is shown in Fig. 2.

1) Main Pipeline: In main pipeline, first, it can be assumed that the last deep feature map $\boldsymbol{M}_{\mathrm{DF}}$ goes through the *softmax* layer and gets the logits map $\boldsymbol{M}_{\mathrm{LG}} \in (H, W, N)$, where N is the number of categories. After that, we perform argmax on the $\boldsymbol{M}_{\mathrm{LG}}$ to obtain the coarse segmentation result $\boldsymbol{O}_C \in (H, W)$

$$\boldsymbol{O}_c = \arg\max_{N} \ (\mathrm{softmax}(\boldsymbol{M}_{\mathrm{DF}})). \tag{8}$$

Calculate the coarse loss value $\mathcal{L}_c$ of the result by applying off-the-shelf loss function. And then update the previous network parameters using chain rule.

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So far this is the traditional way to use DL algorithms for image segmentation. To compensate for the fact that the mainstream deep neural-network-based methods for remote sensing image segmentation allocate all the computational resources equally to each point, we intentionally bias the computational load of the network to those points that are still not well considered by implementing the next two parts, namely, *levity head* (LH) and *deliberation phase* (DP), and then used the considerate points to supplant them in $\boldsymbol{O}_c$ through replacement $\mathcal{R}$. By doing so, those points that deserve more computational resources at the edge can be computed more accurately to improve the overall accuracy of the final image segmentation.

2) Levity Head: LH is proposed here to determine whether enough calculations have been done for each point. We calculate the entropy of all the points in the N directions for each of them

$$\begin{aligned}\boldsymbol{M}_{\mathrm{ET}} &= \mathrm{Ent}(\boldsymbol{M}_{\mathrm{LG}}) \\ &= -\sum_{n=1}^{N} \boldsymbol{M}_{\mathrm{LG}}^{n} \cdot \log\left(\boldsymbol{M}_{\mathrm{LG}}^{n}\right)\end{aligned} \tag{9}$$

and obtain an entropy map $\boldsymbol{M}_{\mathrm{ET}} \in (H, W)$. As entropy is a measure of the amount of information, it means that if the amount of information at a point is greater, the deep network is more uncertain about that point, and vice versa. And according to *information theory*, the entropy of a point on $\boldsymbol{M}_{\mathrm{LG}}$ is maximum when each class of probabilities of it is $(1/N)$, which can be obtained by maximizing the (9) under the restriction

$$\text{s.t.} \sum_{n=1}^{N} \boldsymbol{M}_{\mathrm{LG}}^{n} = \boldsymbol{J}_N \tag{10}$$

using *Lagrange Multipliers*. It is a quiet practical distribution in the classification of a point. For example, assume the logits vector obtained after softmax is $[s_0, s_1, s_2, \ldots, s_N]$. If $s_0 = s_1 = s_2 = \cdots = s_N = (1/N)$, the entropy value of a point in $\boldsymbol{M}_{\mathrm{ET}}$ reaches its maximum value $\log(n)$, which means the classifier is completely unable to discriminate the categories at that point. In contrast, when $s_0 \gg s_1 \approx s_2 \approx \cdots \approx s_N$, it is easy to classify the point into the first category. In this case, the entropy value is much lower than the former situation.

Since levity points do not appear independently and have a strong correlation with their adjacent pixels, after obtaining $\boldsymbol{M}_{\mathrm{ET}}$, a four-way integration operator $\boldsymbol{\mathcal{G}}$ is implemented to filter out these independently occurring singularities while leveraging the ambient information of each point. $\boldsymbol{\mathcal{G}}$ is derived from the classical *Gaussian filter* restricted by a coefficient ξ

$$\xi = \sigma(x+y) + \sigma(x-y) + \sigma(x) + \sigma(y). \tag{11}$$

Let $g(x, y)$ denote the *Gaussian filter* and the elements of operator $\boldsymbol{\mathcal{G}}$ can then be obtained by the following equation:

$$\begin{aligned}g'(x,y) &= \xi \cdot g(x,y) \\ &= \left[\sigma(x+y) + \sigma(x-y) + \sigma(x) + \sigma(y)\right] \cdot e^{\frac{x^2+y^2}{2\sigma^2}}.\end{aligned} \tag{12}$$

Based on $g'(x, y)$, operator $\boldsymbol{\mathcal{G}} \in \mathbb{R}^{2K+1,2K+1}$, where K is the stretch range of the operator, can be constructed as follows:

$$\boldsymbol{\mathcal{G}} = \begin{bmatrix} g'_{(-K,K)} & g'_{(-K+1,K)} & \cdots & g'_{(K,K)} \\ g'_{(-K,K-1)} & g'_{(-K+1,K-1)} & \cdots & g'_{(K,K-1)} \\ \vdots & \vdots & \ddots & \vdots \\ g'_{(-K,-K)} & g'_{(-K+1,-K)} & \cdots & g'_{(K,-K)} \end{bmatrix}. \tag{13}$$

Note that $\boldsymbol{\mathcal{G}}$ reaches the maximum response at the center, namely, $g(0, 0)$, and the response gets lower along the four directions of horizontal, vertical, 45°, and 135°, leaving other position as 0. $\boldsymbol{\mathcal{G}}$ makes $\boldsymbol{M}_{\mathrm{ET}}$ more responsive to levity points at the edges of the feature, and thus enabling contextual information to be extracted from this point. After convolution with $\boldsymbol{\mathcal{G}}$ and then passing a unit step $\mathcal{U}$, we have the levity map $\boldsymbol{M}_{\mathrm{LT}}$

$$\boldsymbol{M}_{\mathrm{LT}} = \mathcal{U}(\boldsymbol{M}_{\mathrm{ET}} * \boldsymbol{\mathcal{G}}). \tag{14}$$

3) Deliberation Phase: In the third part, namely, the DP, the levity points in $\boldsymbol{M}_{\mathrm{LT}}$ are learned further. For these levity points, using only their deep semantic information $\boldsymbol{M}_{\mathrm{DF}}$ is completely inadequate, since they are usually found at the edges of features or in places of relatively narrow scale. The shallow texture features here are an important factor for their classification, and the adjacent information at different scales also has a strong influence on the results. Therefore, we perform feature fusion on these points and its neighbors by the module Fusion ($\mathcal{F}_{\theta_f}$) and Sampler ($\mathcal{S}$). $\mathcal{F}$ fuses the deep semantic feature $\boldsymbol{M}_{\mathrm{DF}}$ with the lower level features $\boldsymbol{M}_{\mathrm{SF1}}$, $\boldsymbol{M}_{\mathrm{SF2}}$, and $\boldsymbol{M}_{\mathrm{SF3}}$ at different scales and depths from the lower triangle (module). Then $\mathcal{S}$ samples the comprehensive feature of the points and its context from the fused feature map according to $\boldsymbol{M}_{\mathrm{LT}}$

$$\boldsymbol{I}_p = \mathcal{S}\left(\mathcal{F}_{\theta_f}(\boldsymbol{M}_{\mathrm{SF1}}, \boldsymbol{M}_{\mathrm{SF2}}, \boldsymbol{M}_{\mathrm{SF3}}, \boldsymbol{M}_{\mathrm{DF}}), \boldsymbol{M}_{\mathrm{LT}}\right). \tag{15}$$

The fused features $\boldsymbol{I}_p$ are then further learned through classifier $\mathcal{C}_{\theta_c}$, making the feature more comprehensive

$$\boldsymbol{O}_p = \mathcal{C}_{\theta_c}(\boldsymbol{I}_p) = [p(n|\theta_c, \boldsymbol{I}_p)], \quad (n = 1, \ldots, N). \tag{16}$$

The classifier $\mathcal{C}$ provides a more deliberate prediction on levity points $\boldsymbol{O}_p$. Correspondingly, the label of the levity points $\boldsymbol{L}_p$ is obtained through $\mathcal{S}$.

With output $\boldsymbol{O}_p$ and label $\boldsymbol{L}_p$ available, the loss value of EIRI, denoted as $\mathcal{L}_e$, can be obtained by any multiclassification loss function

And the definitive loss $\mathcal{L}$ of the method is

$$\mathcal{L} = \mathcal{L}_c + \mathcal{L}_e. \tag{17}$$

$\mathcal{L}_e$ in $\mathcal{L}$ provides a increment gradient Grad_e when updating the front parameters

$$\mathrm{Grad}_e = \frac{\partial \mathcal{L}_e}{\partial \boldsymbol{M}_X} = \theta_c \cdot \theta_f \cdot \frac{\partial \mathcal{L}_e}{\partial \boldsymbol{O}_p}. \tag{18}$$

In (18), $\boldsymbol{M}_X$ denotes one of $\boldsymbol{M}_{\mathrm{SF1}}$, $\boldsymbol{M}_{\mathrm{SF2}}$, $\boldsymbol{M}_{\mathrm{SF3}}$, and $\boldsymbol{M}_{\mathrm{DF}}$. θ_c and θ_f are two learnable parameters which make the network have partiality for those levity point and edge in particular. In backpropagation, Grad_e not only flow to the upper triangle as an auxiliary for the vanilla gradient along

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the chain but also directly stretch into the lower triangle and adjust their parameter directly, which improves the low-level feature saliency.

After LH and DP, the entire algorithm's resource consumption is more inclined to those points that are difficult to classify in the deep network. In this way, the limited computational resources are efficiently and reasonably used.

IV. Experiments

A. Datasets

To evaluate our model, we use four public datasets, Vaihingen data and Potsdam data from the ISPRS dataset,[1] LoveDA [46], and UAVid [47].

The data obtained from Vaihingen, a city in Germany, contains 33 patches of different sizes, the largest of which is a remote sensing image with a size of 3816 × 2550 pixels. It contains both true orthophoto (TOP) and digital surface model (DSM) formats, both as three channels RGB images. The sampling distance of the ground in each image is 9 cm. Potsdam is also located in Germany, and the data obtained from it contain 38 patches. However, unlike Vaihingen data, Potsdam data have the same size of 6000 × 6000 pixels and the ground is sampled at a distance of 5 cm. This dataset provides IRRG (three channels IR-R-G), RGB (three channels R-G-B), and RGBIR (four channels R-G-B-IR) in three different channel combinations. Both the datasets contain six categories, which are impervious surfaces, building, low vegetation, tree, car, and clutter/background.

LoveDA is a ground cover dataset open-sourced by the RSIDEA team at Wuhan University. LoveDA's data are derived from 5987 remote sensing images with 0.3-cm resolution from three cities in China. Since the dataset contains urban and rural areas, it causes multiscale feature and variable samples. UAVid is a high-resolution semantic segmentation dataset comprising 30 video sequences captured by UAVs. Therefore, this dataset increases the difficulty of moving target recognition and segmentation. These multiscale features also become a significant influence at different shooting heights.

B. Implement Details

1) Parameter Setting: For all the datasets, we use SGD as the optimizer. For Vaihingen, Potsdam, and LoveDA, the batch size for training and testing is set to 8. For UAVid, it is set to 4. We use a learning rate of e-4. The number of boundary points $\boldsymbol{k}$ is set to 3000, and the number of eigenvalues is set to 2.

2) Experimental Configurations: In this article, we use Workstation in our experiments, which contains 64G RAM and four Geforce RTX 2080Ti GPUs, each containing 11 GB of memory. The system version is Ubuntu 16.04 LTS, the programming language is Python 3.6.8, and the framework is PyTorch 1.2.0. To ensure that the model is fair for both the datasets, we use three-channel RGB images in each of the following segmentation experiments.

[1]https://www2.isprs.org/commissions/comm2/wg4/benchmark/2d-semlabel-potsdam/

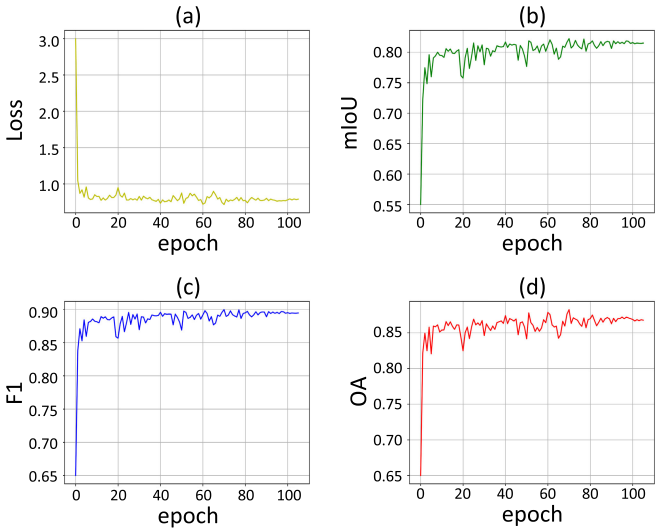


Fig. 3. (a) Curve of loss, (b) curve of mIoU, (c) curve of F_1 score, and (d) curve of OA in Potsdam.

C. Evaluate Metrics

We use F_1 and overall accuracy (OA) to evaluate our method, and they can be calculated as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}$$
$$\mathrm{F}_1 = 2 \times \left(\frac{P \cdot R}{P+R}\right)$$
$$\mathrm{OA} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}+\mathrm{TN}} \tag{19}$$

where TP, FP, TN, and FN are the true positives, false positives, true negatives, and false negatives, respectively, P represents precision, and R represents recall.

For Potsdam, Fig. 3 shows the variation curves of loss, mIoU, F_1 score, and OA with epoch. It can be seen that after five epochs, the metrics are at a high level and gradually stabilize afterward. This indicates that our method is effective and can converge to high accuracy.

D. Ablation Study

This part will use abundant ablation experiments to verify the improvement effect of each module on the segmentation results. We use HRNet as the baseline, rebuild FEN, and add SPN to form the base network of C-PNet. Then ICEN and CPB are added in turn. In our experiments, we use F_1 scores and OA to evaluate the performance of the model on the ISPRS dataset under each stage.

1) Performance Improvement of ICENet: We compare the segmentation results of HRNet and C-PNet for small targets under the same conditions. In Fig. 4, we can see that C-PNet ensures the segmentation accuracy of HRNet for large targets (e.g., impervious surfaces and buildings) while enabling better segmentation of small targets (e.g., low vegetation and car) as well.

2) Effect of CPB: The high-level semantic features have a strong guiding effect on the low-level semantic features. For example, the edges in the low vegetation, which has a strong

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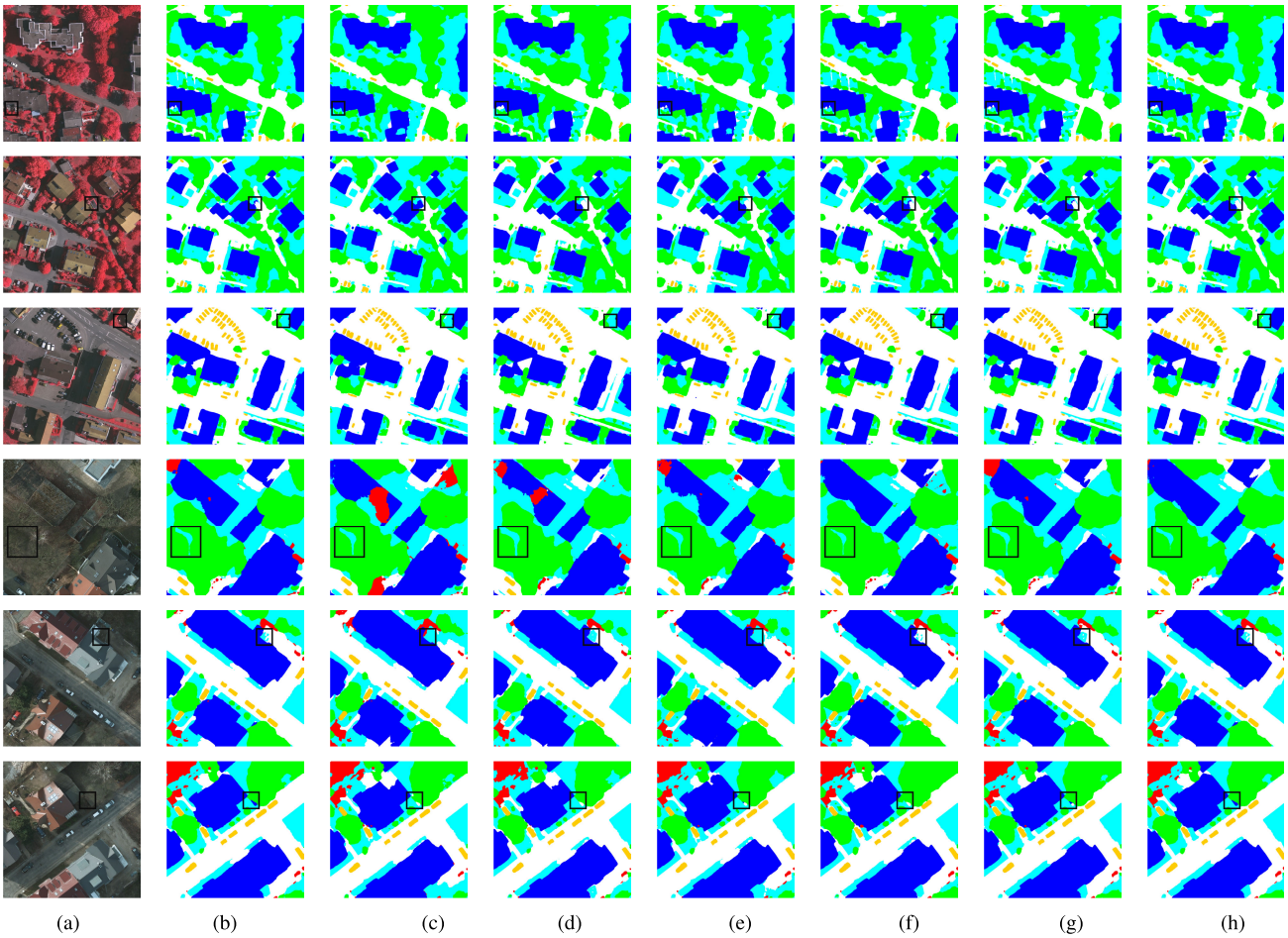


Fig. 4. Qualitative comparison over five methods on the ISPRS 2-D Semantic labeling Vaihingen dataset (upper three rows) and Potsdam dataset (lower three rows). From left to right are (a) input image, (b) ground truth, (c) UZ, (d) UFMG_4, (e) RIT_L7, (f) AFNet, (g) HRNet, and (h) C-PNet, respectively. Input images are cropped into 512 × 512 and other hyperparameters are kept the same. The relatively obvious areas of improvement are marked with black rectangles.

variation in texture features. Based on this, it is possible to have an obvious guiding effect on the shape of low vegetation. As shown in Fig. 4, it can be seen that when two branches share feature information through a gating function and use high-level features to control whether the low-level features are fused, it will give better segmentation results for the region with more categories.

3) EIRI: The chariness decision module improves the confidence of the neural network by reconsidering the levity points which aims to alleviate the errors in the edge and small targets. Experiments on different datasets are performed to validate the effectiveness of the chariness decision module. Fig. 4 gives a distinct visual comparison in terms of edge segmentation results between HRNet and HRNet + CD and the levity points in HRNet. Without the chariness decision module, HRNet generates amount of levity points causing the output land cover map exhibiting a lumpy shape even though the ground target contains many angular constructions. From the figure, we can tell that most of the levity points locate at the edge of the targets. After DP, these points obtain more computational incline and thus yield a better performance. Table I numerically represents the effectiveness of the chariness decision module by calculating two matrices, F_1 score and OA, respectively, on the class with prominent angular border. Thanks to the improvement on the edge, their value exhibits a more satisfactory result especially for the urban scene.

TABLE I
PERFORMANCE COMPARISON BASED ON THE POTSDAM DATASET

Method	F_1	**OA**
Baseline	90.3	89.3
ICEN	90.3	89.5
ICEN + CPB	91.0	89.8
ICEN + CPB + EIRI	91.5	90.1

E. Comparison to Other Methods

In this section, the proposed method is compared with other five methods on the ISPRS 2-D semantic labeling Vaihingen and Potsdam datasets, LoveDA, and UAVid. The input image is cropped into 512 × 512 and other parameter is also kept the same to ensure the experimental fairness. The results

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TABLE II

QUANTITATIVE COMPARISON OVER 11–15 METHODS ON THE ISPRS 2-D SEMANTIC LABELING VAIHINGEN DATASET AND POTSDAM DATASET. THE VALUES IN BOLD ARE THE BEST SCORES AMONG THESE METHODS AND UNDERLINED ONES ARE THE SECOND BEST

Dataset	Method	Im. Surf. (%)	Build. (%)	L. Veg. (%)	Tree (%)	Car (%)	F_1 scores	OA
Vaihingen	DABNet	87.8	88.8	74.3	84.9	60.2	79.2	84.3
	ERFNet	88.5	90.2	76.4	85.8	53.6	78.9	85.8
	BiSeNet	89.1	91.3	80.9	86.9	73.1	84.3	87.1
	UZ	89.2	92.5	81.6	86.9	57.3	81.5	87.3
	PSPNet	89.0	93.2	81.5	87.7	43.9	79.0	87.7
	BoTNet	89.9	92.1	81.8	88.7	71.3	84.8	88.0
	Segmenter	89.8	93.0	81.2	88.9	67.6	84.1	88.1
	DANet	90.0	93.9	82.2	87.3	44.5	79.6	88.2
	FANet	90.7	93.8	82.6	88.6	71.6	85.4	88.9
	UFMG_4	91.1	94.5	82.9	88.8	81.3	-	89.4
	EaNet	91.7	94.5	83.1	89.2	80.0	87.7	89.7
	ShelfNet	91.8	94.6	83.8	**89.3**	77.9	87.5	89.8
	MAResU-Net	92.0	95.0	83.7	**89.3**	78.3	87.7	90.1
	SwiftNet	92.2	94.8	84.1	**89.3**	81.2	88.3	90.2
	C-PNet	**92.4**	**95.1**	**84.2**	89.1	**82.0**	**88.7**	**90.5**
Potsdam	UZ	89.3	95.4	81.8	80.5	86.5	86.7	85.8
	DeepLabv3+	88.7	95.0	82.0	81.7	83.8	88.1	86.2
	PSPNet	89.5	95.0	82.8	82.6	85.6	88.8	87.1
	U-Net	88.8	93.8	83.2	83.4	88.4	89.0	87.5
	UFMG_4	90.8	95.6	84.4	84.3	92.4	-	87.9
	RIT_L7	91.2	94.6	**85.1**	85.1	**92.8**	89.8	88.4
	U-Net-MS	90.0	95.1	84.5	85.7	89.0	**91.5**	88.9
	HRNet-OCR	90.7	95.2	84.5	85.8	90.6	90.3	89.3
	HRNet	91.5	95.7	84.8	85.9	90.3	90.5	89.6
	EfficientPS	89.2	92.9	83.5	84.4	87.1	87.4	90.0
	C-PNet	**92.1**	**95.8**	83.8	**86.8**	91.3	**91.5**	**90.1**

of qualitative and quantitative comparison over the 11–15 methods are shown in Fig. 4 and Table II, respectively.

1) Results on Vaihingen Dataset: First, we conduct experiment on the ISPRS 2-D semantic labeling Vaihingen dataset. The upper half part of Table II details the numerical results of the six methods on this dataset. UZ exploits convolution to learn semantic features from the input image and then performs deconvolution to upsample the concentrated high-level features to the original resolution. In this way, the model gives a dense prediction of every pixel on the original input image. However, UZ has lost amount of detailed features as convolution significantly expands the receptive field, and therefore yields poorer performance than that of others.

RIT_L7 improves the performance from another point of view; in high level, they argue to fuse the multisensor data from remote optical device and LiDAR by a decision-level fusion strategy achieved by a higher order conditional random field (CRF) and produce a result instructed by these data obtained from multiple devices. However, it is still suffering from the deficiency of low-level features. UFMG_4 harnesses dilated convolution to extract multicontext features of different scales and enrich more abundant features, alleviating the lack of spatial semantic information. And AFNet autonomously performs multiscale fusion operation based on the size of the perceptual field inspired by nonlocal neural network [45]. Thanks to its adaptively adjusted fusion strategy, the effect of the size of the targets no longer limits the network's learning ability so severely, and it provides a significantly increased performance on all the categories especially on low vegetables.

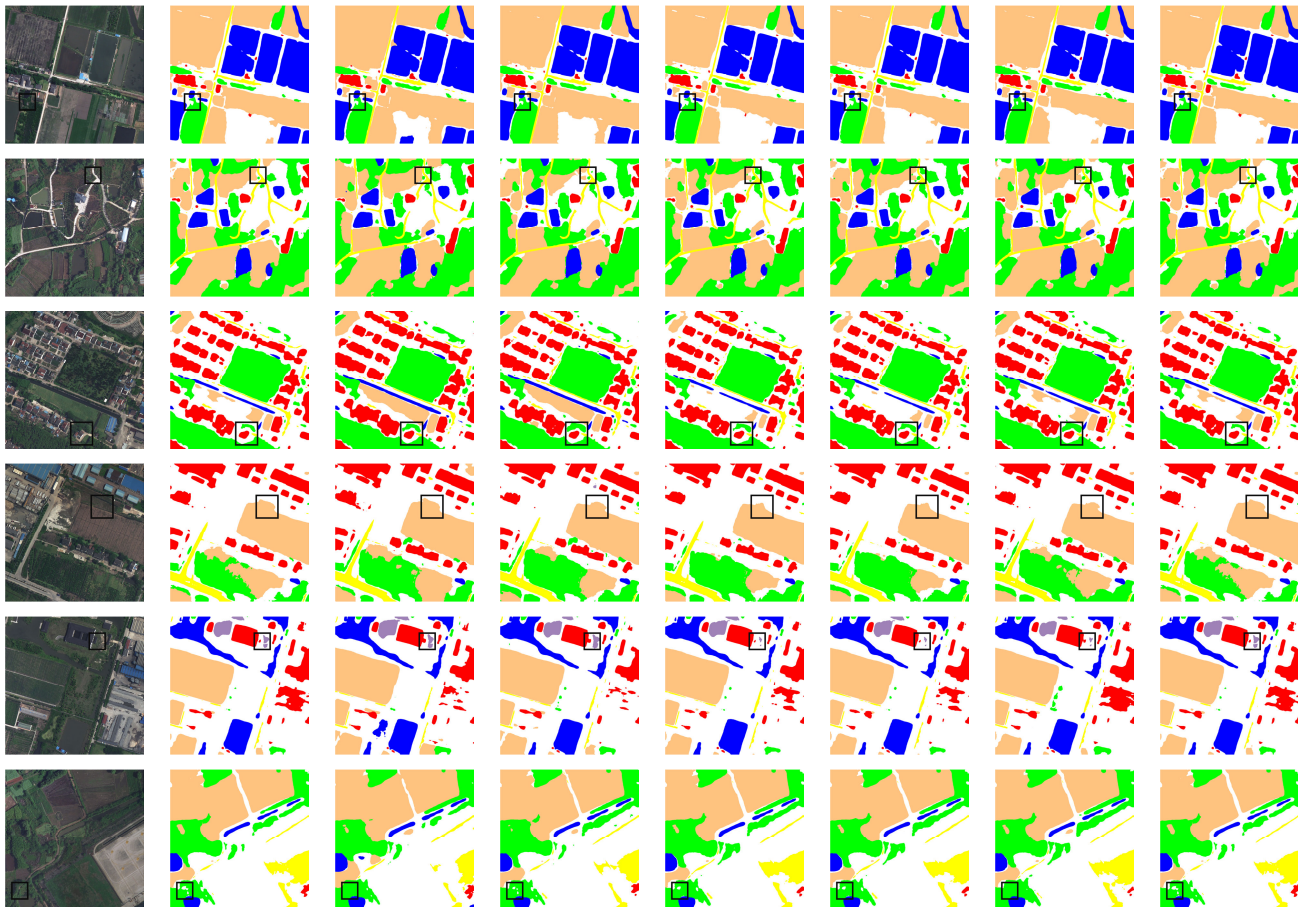
Compared with our proposed method, however, they either neglect too much low-level features which facilitate target fine-edge description nor call for data obtained from other devices to contemplate the very-high-resolution aerial optical image. The proposed C-PNet yields more satisfying results either visually or numerically and achieves the highest accuracy in the category of building, impermeable surface, and car. With the help of ICENet and EIRI, more detailed features with different spatial resolution are able to be maintained and finer grained features can be further learned compared with other four DL method, and therefore it exhibits more detailed ground targets. From the results, it can also be observed that the outline of the land cover map is more sharp and clings to the targets' edges because of the equitable calculational resource allocation and the competence of shallow feature controlled by CPB.

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TABLE III

PERFORMANCE COMPARISON BASED ON THE LOVEDA DATASET. THE VALUES IN BOLD ARE THE BEST SCORES AMONG THESE METHODS AND UNDERLINED ONES ARE THE SECOND BEST

Method	Background	Building	Road	Water.	Barren	Forest	Agriculture	mIoU
Segmenter	38.0	50.7	48.7	77.4	13.3	43.5	58.2	47.1
DeepLabv3+	43.0	50.9	52.0	74.4	10.4	44.2	58.5	47.6
PSPNet	**44.4**	52.1	53.5	76.5	9.7	44.1	57.9	48.3
TransUNet	43.0	**56.1**	53.7	78.0	9.3	44.9	56.9	48.9
BANet	43.7	51.5	51.1	76.9	**16.6**	44.9	**62.5**	49.6
C-PNet	44.0	55.2	**55.3**	**78.8**	16.0	**46.4**	58.0	**51.8**



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 5. Qualitative comparison over five methods on the LoveDA dataset. From left to right are (a) input image, (b) ground truth, (c) PSPNet, (d) DeepLabv3+, (e) BANet, (f) TransUNet, (g) Segmenter, and (h) C-PNet, respectively. Input images are cropped into 512 × 512, and other hyperparameters are kept the same. The relatively obvious areas of improvement are marked with black rectangles.

2) Results on Potsdam Dataset: The experiment results are shown in Table I, from which we can see that the segmentation accuracy of HRNet in the ISPRS dataset is already competitive. Meanwhile, we can find its generalization ability in different resolution remote sensing images. The ISPRS 2-D semantic labeling Potsdam dataset exhibits similar results to the Vaihingen dataset. UFMG_4 and RIT_L7 achieve comparable performance, and both have a remarkable improvement compared with UZ, but UZ shows a slight lead in class building over RIT_L7. This is because the buildings in Potsdam region arrange more regularly and therefore reduce learning complexity. In other words, although RIT_L7 collects data from various sources, much redundant information exists compared with raw RGBIR optical data. After adding SPN and CPB, the accuracy of each category is improved, except for low vegetation, and the F_1 scores and OA are increased by 0.7 and 0.5 points, respectively. The more accurate contouring of feature edges confirms the ability to improve the edge processing accuracy without increasing the computational effort.

3) Results on LoveDA Dataset: We conduct experiments on the LoveDA dataset, which is used to verify the effectiveness of C-PNet. Table III shows the comparison results with five methods, and the visualized results are shown in Fig. 5. As can be seen from the results in the black rectangle, EIRI

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TABLE IV
PERFORMANCE COMPARISON BASED ON THE UAVID DATASET. THE VALUES IN BOLD ARE THE BEST SCORES AMONG THESE METHODS AND UNDERLINED ONES ARE THE SECOND BEST

Method	Building	Tree	Clutter	Road	L. Veg.	S. Car	M. Car	Human	mIoU
FCN-8s	64.3	63.8	33.5	57.6	28.1	8.4	29.1	0.0	35.6
Dilation	72.8	66.9	38.5	62.4	34.4	1.2	36.8	0.0	39.1
U-Net	70.7	67.2	36.1	61.9	32.8	11.2	47.5	0.0	40.9
MSD	**80.9**	75.5	46.3	66.7	47.9	22.3	56.9	4.2	50.1
MANet	79.2	72.5	53.1	**74.3**	53.7	30.8	56.6	28.6	56.6
C-PNet	60.5	**79.3**	**73.5**	72.3	**55.4**	**49.8**	**66.5**	**29.0**	**60.1**

can help the model to be able to better segment complex and variable scene areas in a complex and variable scene. In particular, C-PNet also achieved competitive segmentation in areas where the low-margin vegetation edge feature classes are not prominent.

4) Results on UAVid Dataset: We conduct experiments on the LoveDA dataset, which is used to verify the effectiveness of C-PNet. Table IV shows the comparison results with five methods. Since drones capture UAVid's data, the same category does not have the same size and resolution in different images. Compared with competing algorithms U-Net, MSD, and MANet, C-PNet improves the segmentation accuracy of small targets. In particular, for the car, the mIoU of C-PNet can be improved by ten points. It proves that ICEN has good feature extraction capability for different resolution features.

F. Discussion

The improvement of the results requires more computational resources compared with other deep-learning-based methods. Therefore, it takes a longer time to predict the land cover map. It can be inferred that ICEN occupies a majority of the excess parameters. We solve this problem by making HRNet more sparse to pay more attention to deep semantic features while adaptively fusing the feature by calculating a gate and avoiding redundant parameters. Using these strategies, our method does not burden the computational load severely and reaches a decent result. For a fair comparison, the input images are all cropped to the size of 512 $\times$ 512. In practice, it can be a random size for the input image as there are no fully connected layers in our network, which is of great necessity when meeting the large aerial land cover mapping task.

V. CONCLUSION

This article proposes a semantic network for targets of different sizes. Combining FEN with SPN enables targets of different sizes to be well segmented. CPB uses high-level semantic features to control whether low-level semantic features are fused to deeper networks, which ensures that low-level features are retained in deeper networks while reducing the influence of label noise in low-level features. EIRI enables further segmentation of uncertain locations of target edges without consuming too much computation on regions within a single target where the category has already been determined. However, more research is needed to demonstrate its effectiveness for the potential of CPB. Meanwhile, how C-PNet can effectively use the features in the NIR band still requires future research.

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